## SQL Code

```sql
USE waitlist_system;

DROP TABLE IF EXISTS waitlist_log;
DROP TABLE IF EXISTS notification;
DROP TABLE IF EXISTS waitlist_entry;
DROP TABLE IF EXISTS status;
DROP TABLE IF EXISTS staff;
DROP TABLE IF EXISTS customer;

CREATE TABLE customer (
	customer_id INT PRIMARY KEY AUTO_INCREMENT,
	first_name VARCHAR(100) NOT NULL,
	last_name VARCHAR(100) NOT NULL,
	phone_number VARCHAR(20) NOT NULL,
	email VARCHAR(100),
	preferred_contact_method VARCHAR(10) NOT NULL
		CHECK (preferred_contact_method IN ('Text', 'Email'))
);

CREATE TABLE staff (
	staff_id INT PRIMARY KEY AUTO_INCREMENT,
	first_name VARCHAR(100) NOT NULL,
	last_name VARCHAR(100) NOT NULL,
	username VARCHAR(20) NOT NULL UNIQUE,
	password_hashed VARCHAR(255) NOT NULL
);

CREATE TABLE status (
	status_id INT PRIMARY KEY AUTO_INCREMENT,
	status_name VARCHAR(20) NOT NULL UNIQUE
);

CREATE TABLE waitlist_entry (
	entry_id INT PRIMARY KEY AUTO_INCREMENT,
	customer_id INT NOT NULL,
	staff_id INT NOT NULL,
	status_id INT NOT NULL,
	party_size INT NOT NULL,
	requested_reservation_time DATETIME,
	check_in_type VARCHAR(10) NOT NULL
		CHECK (check_in_type IN ('Walk-in', 'Text', 'Web-app')),
	notes TEXT,
	created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,

	FOREIGN KEY (customer_id) REFERENCES customer(customer_id),
	FOREIGN KEY (staff_id) REFERENCES staff(staff_id),
	FOREIGN KEY (status_id) REFERENCES status(status_id)
);
```

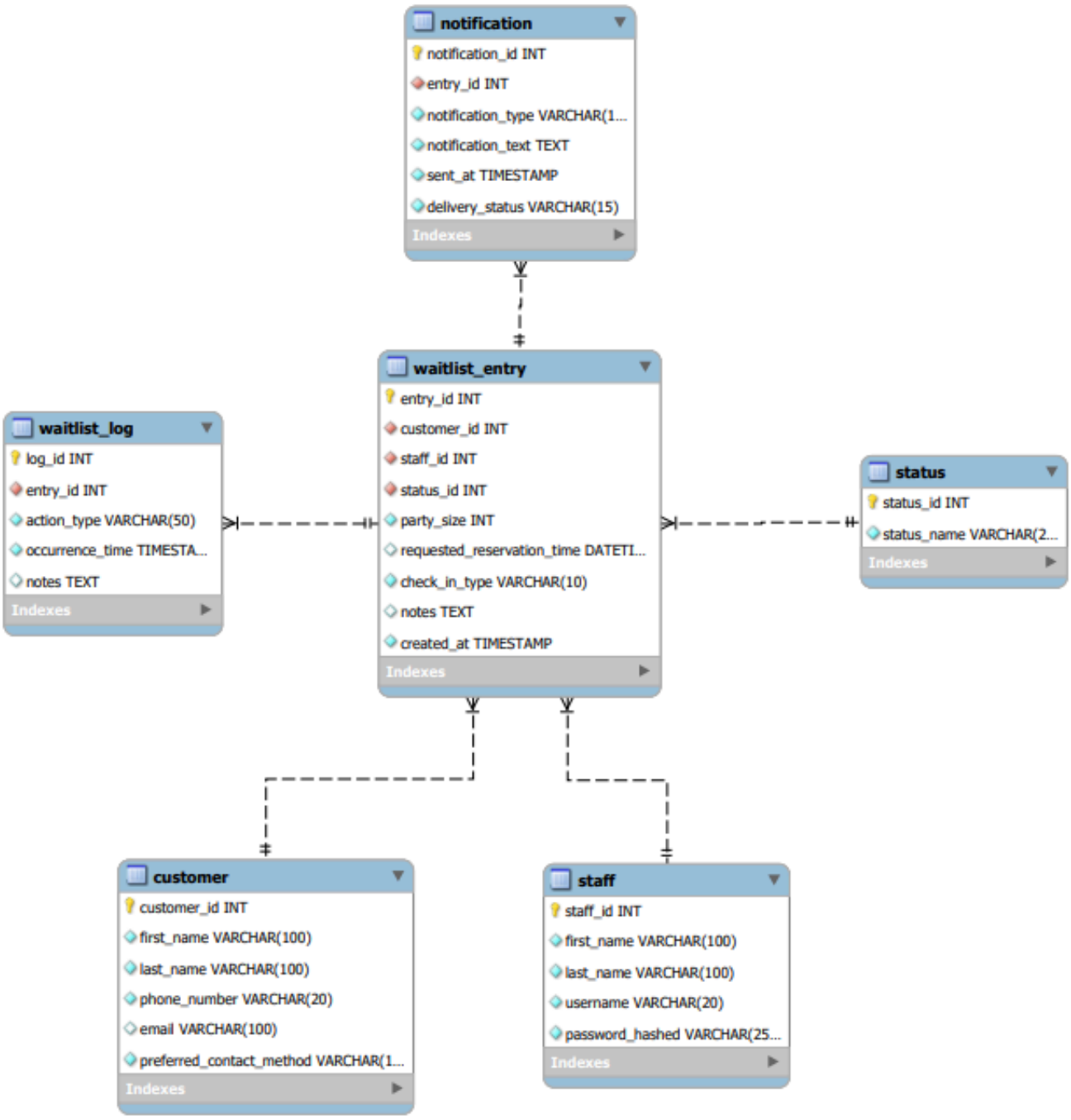
```sql
CREATE TABLE notification (
	notification_id INT PRIMARY KEY AUTO_INCREMENT,
	entry_id INT NOT NULL,
	notification_type VARCHAR(10) NOT NULL
		CHECK (notification_type IN ('Text', 'Email')),
	notification_text TEXT NOT NULL,
	sent_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
	delivery_status VARCHAR(15) NOT NULL
		CHECK (delivery_status IN ('Sent', 'Not-delivered')),

	FOREIGN KEY (entry_id) REFERENCES waitlist_entry(entry_id)
);

CREATE TABLE waitlist_log (
	log_id INT PRIMARY KEY AUTO_INCREMENT,
	entry_id INT NOT NULL,
	action_type VARCHAR(50) NOT NULL,
	occurrence_time TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
	notes TEXT,

	FOREIGN KEY (entry_id) REFERENCES waitlist_entry(entry_id)
);
```

**ERD Diagram**


notification
notification_id INT
entry_id INT
notification_type VARCHAR(1...
notification_text TEXT
sent_at TIMESTAMP
delivery_status VARCHAR(15)
Indexes
waitlist_entry
entry_id INT
customer_id INT
staff_id INT
status_id INT
party_size INT
requested_reservation_time DATETI...
check_in_type VARCHAR(10)
notes TEXT
created_at TIMESTAMP
Indexes
waitlist_log
log_id INT
entry_id INT
action_type VARCHAR(50)
occurrence_time TIMESTA...
notes TEXT
Indexes
status
status_id INT
status_name VARCHAR(2...
Indexes
customer
customer_id INT
first_name VARCHAR(100)
last_name VARCHAR(100)
phone_number VARCHAR(20)
email VARCHAR(100)
preferred_contact_method VARCHAR(1...
Indexes
staff
staff_id INT
first_name VARCHAR(100)
last_name VARCHAR(100)
username VARCHAR(20)
password_hashed VARCHAR(25...
Indexes

## Screenshots of sample data

### customer table

| customer_id | first_name | last_name | phone_number | email | preferred_contact_method |
|---|---|---|---|---|---|
| 1 | Dongthi | Nguyen | 4435551234 | dongthi.nguyen@gmail.com | Text |
| 2 | Steven | Zhang | 4105555678 | steven.zhang@gmail.com | Email |
| 3 | Chris | Song | 3015558899 | chris.song@gmail.com | Text |
| 4 | Aaki | Roka | 6675551122 | aaki.roka@gmail.com | Email |
| 5 | Vivian | Tran | 2405553344 | vivian.tran@gmail.com | Text |
| NULL | NULL | NULL | NULL | NULL | NULL |

### staff table

| staff_id | first_name | last_name | username | password_hashed |
|---|---|---|---|---|
| 1 | Kevin | Nguyen | Knguyen | hashed_pw_3 |
| 2 | Vivian | Pham | Vpham | hashed_pw_2 |
| 3 | Kim | Tran | Ktran | hashed_pw_1 |
| NULL | NULL | NULL | NULL | NULL |

### status table

| status_id | status_name |
|---|---|
| 5 | Cancelled |
| 6 | No-Show |
| 3 | Notified |
| 2 | Ready |
| 4 | Seated |
| 1 | Waiting |
| NULL | NULL |

### waitlist_entry table

| entry_id | customer_id | staff_id | status_id | party_size | requested_reservation_time | check_in_type | notes | created_at |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 2 | 2026-04-04 18:30:00 | Walk-in | Window seat preferred | 2026-04-06 00:59:08 |
| 2 | 2 | 1 | 1 | 4 | 2026-04-04 19:00:00 | Web-app | Birthday celebration | 2026-04-06 00:59:08 |
| 3 | 3 | 2 | 2 | 3 | 2026-04-04 18:45:00 | Text | Allergic to peanuts | 2026-04-06 00:59:08 |
| 4 | 4 | 2 | 3 | 5 | 2026-04-04 19:15:00 | Walk-in | NULL | 2026-04-06 00:59:08 |
| 5 | 5 | 3 | 4 | 2 | 2026-04-04 18:20:00 | Web-app | VIP customer | 2026-04-06 00:59:08 |
| NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |

### notification table

| notification_id | entry_id | notification_type | notification_text | sent_at | delivery_status |
|---|---|---|---|---|---|
| 1 | 3 | Text | Your table is ready! | 2026-04-05 23:56:01 | Sent |
| 2 | 4 | Email | Please return to the host stand. | 2026-04-05 23:56:01 | Sent |
| 3 | 2 | Text | You are next in line. | 2026-04-05 23:56:01 | Not-delivered |
| NULL | NULL | NULL | NULL | NULL | NULL |

### waitlist_log table

| log_id | entry_id | action_type | occurrence_time | notes |
|---|---|---|---|---|
| 1 | 1 | Created | 2026-04-05 23:56:01 | Customer added to waitlist |
| 2 | 2 | Created | 2026-04-05 23:56:01 | Reservation made via web |
| 3 | 3 | Status Updated | 2026-04-05 23:56:01 | Changed from Waiting to Ready |
| 4 | 3 | Notification Sent | 2026-04-05 23:56:01 | Customer notified via text |
| 5 | 5 | Seated | 2026-04-05 23:56:01 | Customer has been seated |
| 6 | 4 | Notification Sent | 2026-04-05 23:56:01 | Email notification sent |
| NULL | NULL | NULL | NULL | NULL |